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Team 218 - Global-Therma

Delayed detection of neonatal sepsis leads to preventable infant mortality, lifelong health complications, and rising healthcare costs. Infrared thermography-enabled NICU beds for early detection of neonatal sepsis

Team Introduction

Neo-Therm

Delayed detection of neonatal sepsis leads to preventable infant mortality, lifelong health complications, and rising healthcare costs. Infrared thermography-enabled NICU beds for early detection of neonatal sepsis



NEO-THERM

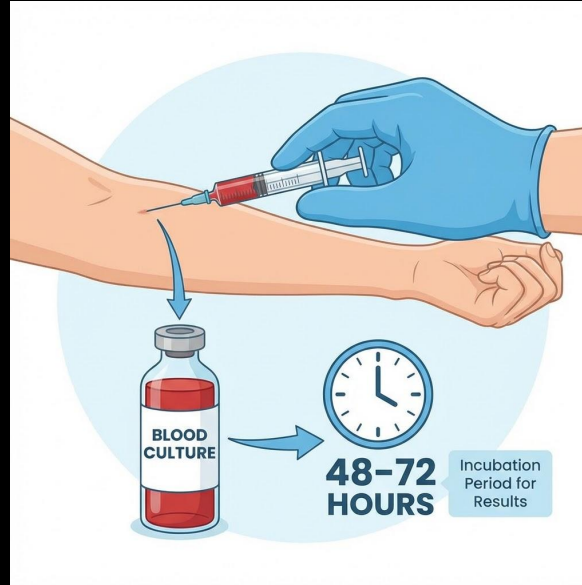
MIT GrandHack

✓ Intersystems Prize



18





Non-Specific
Variable
Late

Before traditional diagnostics detect sepsis, microvascular disruption creates subtle thermal signatures on the skin that can be captured with infrared thermography.

Calculating Market Size



Conventional diagnostic methods



INVASIVENESS



- Blood tests: Lactate, Procalcitonin (PCT), C-reactive protein (CRP), Complete blood count (CBC)
- Lumbar puncture
- Arterial blood gas (ABG)

TIME DELAY



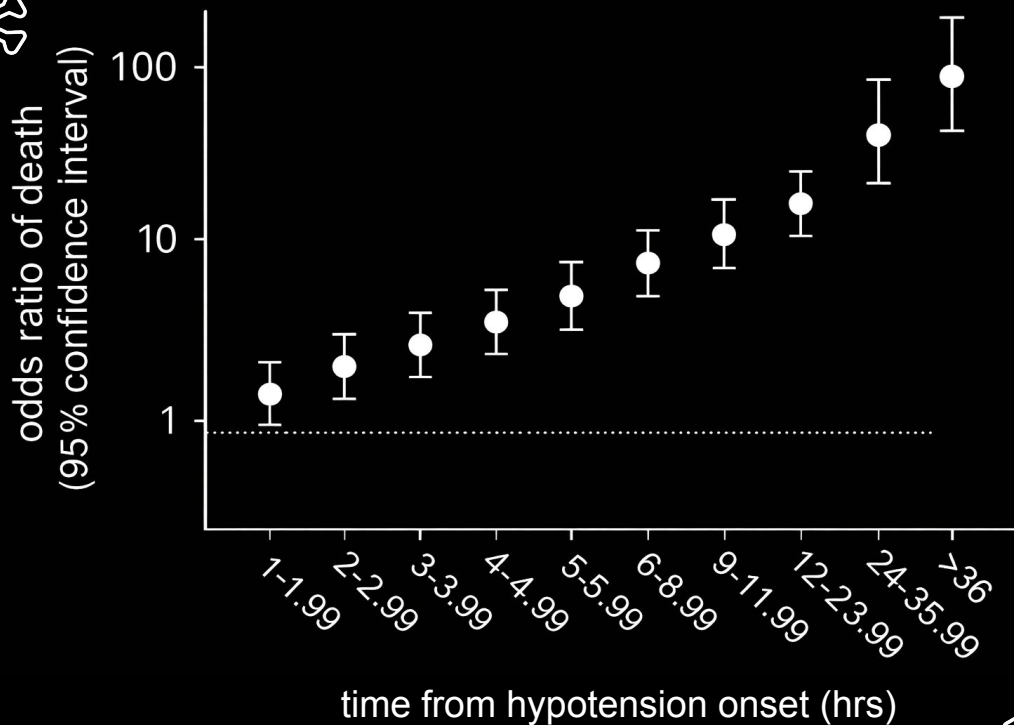
- Scores: Rodwell score, nSOFA, Tollner
- Blood Culture
- Clinical signs: innespecific, late

COMPLEX INFRASTRUCTURE



- Microbiology laboratories (culture incubation systems)
- Automated blood culture systems (BACTEC / BacT-ALERT)
- PCR molecular diagnostic platforms

In sepsis, time is survival.

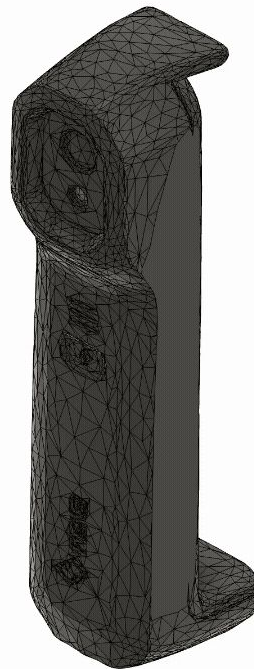


LOGO LOGO LOGO

Thermography

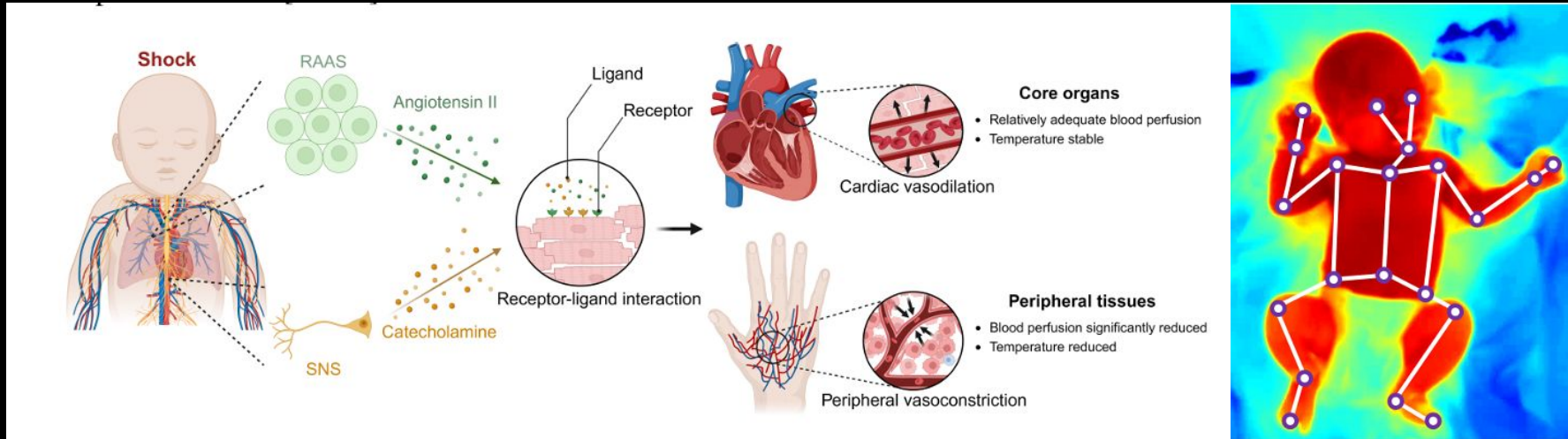
Temperature + AI

- 01** Low-cost FDA approved Camera
- 02** *Thermal Imaging in a highly Temperature Controlled Environment*
- 03** Continuously monitors CPTD (Core-peripheral temperature difference)



How does it work?

Hand vs Trunk



- **Early sepsis alters microcirculation**
- **This creates abnormal temperature differences on the causing abnormal core–peripheral temperature differences**

> [Pediatr Infect Dis J.](#) 2017 Dec;36(12):e293-e297. doi: 10.1097/INF.0000000000001688.

Central-peripheral Temperature Monitoring as a Marker for Diagnosing Late-onset Neonatal Sepsis

José Luis Leante-Castellanos¹, Antonio Martínez-Gimeno, Manuel Cidrás-Pidré, Gerardo Martínez-Munar, Ana García-González, Carmen Fuentes-Gutiérrez

Affiliations + expand

PMID: 28719503 DOI: [10.1097/INF.0000000000001688](#)

io-Remote Sensing in Predicting Infection in Neonates With Thermal Imaging and Machine Learning

April 2022

DOI: [10.1109/SSI56489.2022.9901414](#)

Conference: 2022 Smart Systems Integration (SSI)

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► [Biomed Opt Express.](#) 2025 Feb 10;16(3):965–981. doi: [10.1364/BOE.549693](#)

Thermal imaging-based core peripheral temperature difference measurement for neonatal monitoring in the NICU

[Nantao Zhang](#)¹, [Xiaoyan Song](#)², [Junli He](#)³, [Fengchao Liang](#)¹, [Jie Yang](#)², [Wenjin Wang](#)^{3,*}

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC11919349 PMID: [40109525](#)

Core and peripheral temperature as a marker for sepsis in the preterm infant

Jacqueline Smith (Dr)^a, Gary Alcock (Dr)^b, Barbara Monk NNP^b, Emeritus Rhondda Jones (Professor)^c

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<https://doi.org/10.1016/j.jnn.2021.08.014>

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[1186/s12884-019-2533-y](#)

Thermal imaging applications in neonatal care: a scoping review

[Anastasia Topalidou](#)^{1,✉}, [Nazmin Ali](#)¹, [Slobodan Sekulic](#)², [Soo Downe](#)¹

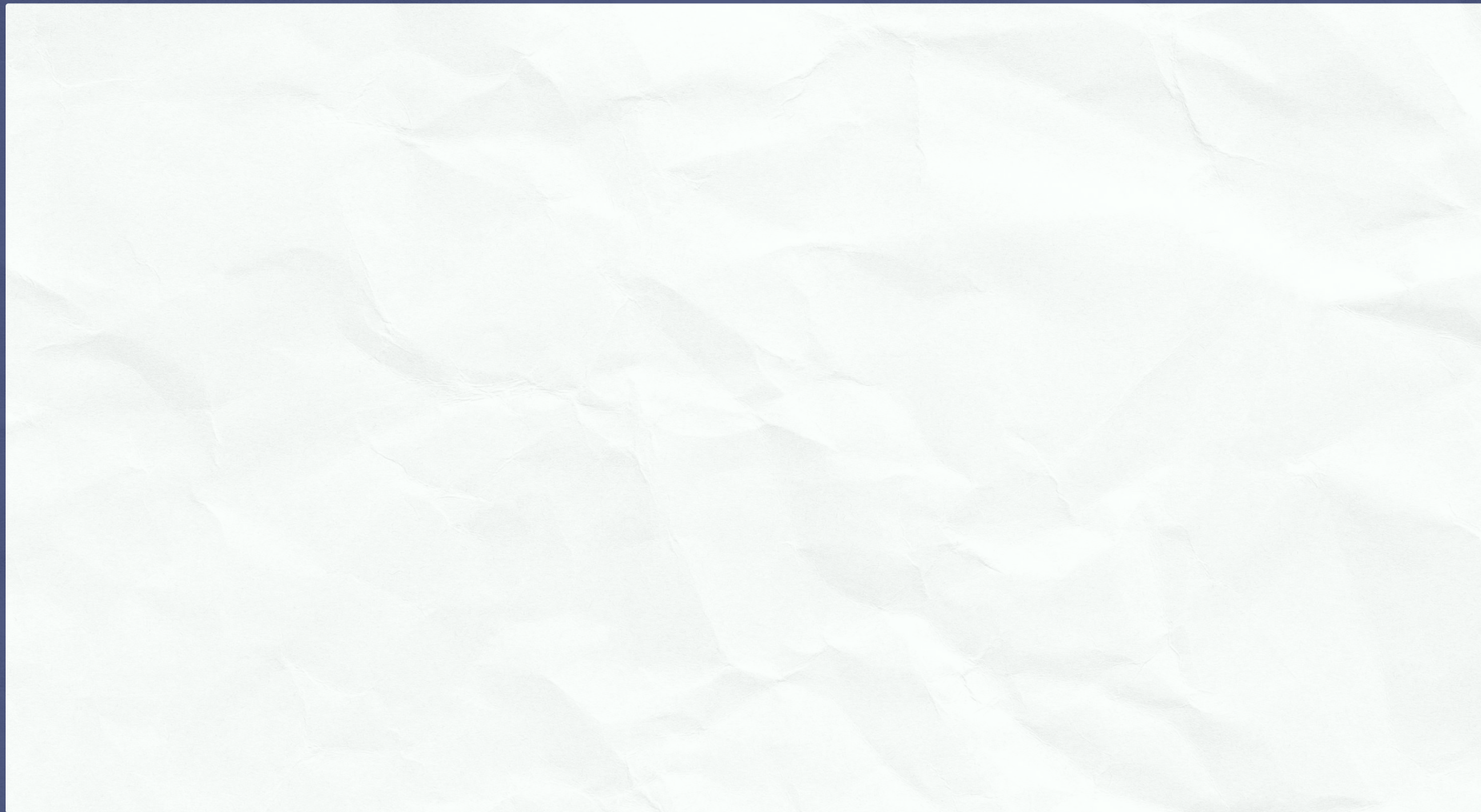
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20 most costly inpatient conditions

Total aggregate hospital costs for 35.6 million hospital stays totaled \$381.4 billion in 2013, with 20 conditions accounting for nearly half of hospital costs, according to a [statistical brief](#) from the Healthcare Cost and Utilization Project.

1. Septicemia—

That immune response, known as sepsis, is the third most common cause of death in U.S. hospitals and [affects 1.7 million people](#) nationwide each year, according to the Centers for Disease Control and Prevention (CDC). It's also the most expensive condition, costing [an estimated \\$62 billion](#) annually in hospitalizations and skilled nursing care.



Competitors

	Continuous	Non Invasive	Neonatal Specific	Early Detection
Prenosis				
Mednition				
Epic Sepsis Model				
En				
H				

Working with Experts in the Field



Weinberger, Barry

Assoc Director • Neonatology at CCMC



Zanos, Theodoros

AVP, Exploration and Innovation • Division of Health AI



Roadmap

Teams

T1/26

T2/26

T1/27

T2/27

Produto

Escrita do projeto e submissão ao comitê de Ética

Pesquisa clínica na Santa Casa de Misericórdia de São Paulo



Parceiro

Apoio Harvard T. H. Chan



HARVARD T.H. CHAN
SCHOOL OF PUBLIC HEALTH

Parceria com Academia Brasileira de Laringologia e Voz

Incubação Google for Startups

Desenvolvimento e Investimento

Fine-tuning com database bridge2ai

Fechamento de POC



MVP com dados clínicos brasileiros

Investimento Pre-Seed

Meet our Team

Uniquely Equipped to Build
Global-Therma



Kieran Tran
(Elec Eng/ Maths)



Ana Clara Rosa
(Medicine)



Sebastián Cicconi (Gene
Editing/Neuroscience)

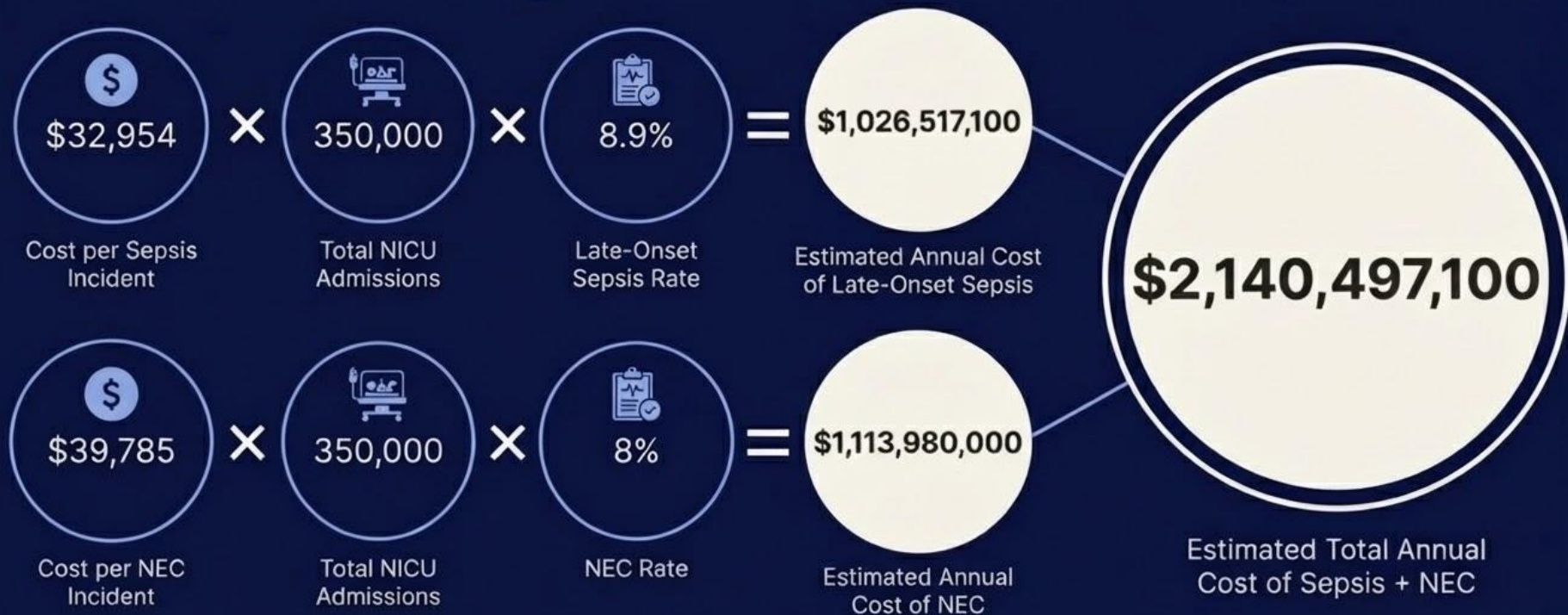
final SENTENCE - impactful



NEO-THERM

Appendix
below

Calculating Market Size



Appendix – System Architecture

NICU Bedside

FLIR A400/A700

Radiometric, 640x480, GigE
Vision
510(k) cleared, product code
LHQ

PHASE 1

Device Middleware

Philips Capsule MDIP

Vendor-neutral device integration
Receives HL7v2 on device VLAN

EXISTING

Integration Engine

InterSystems Health Connect

HL7v2 ↔ FHIR R4 transform
Message queue, retry, audit log
Routes to legacy EHR OR Epic

EXISTING

EHR

Flowsheet

CPTD + zone temps as vital sign
rows

PHASE 1

Polyethylene IR Window

92–94% LWIR transmission
Insert in incubator access port

FHIR R4 Repository

InterSystems IRIS for Health
Queryable Observation history

PHASE 2-3

BestPractice Alert

Hospital-configured rule on CPTD

PHASE 2

Edge Compute Box

IEC 60601-1 certified
Spinnaker SDK + YOLOv8-Pose
CPTD computation, 24h local
buffer

SMART on FHIR Widget

Dedicated NeoTherm tab in chart

PHASE 3

Appendix – HL7v2 Integration (Phase 1)

HL7v2 ORU^R01 Message Format

Edge box emits standard HL7v2 ORU messages on the medical device VLAN. Capsule receives it identically to any bedside monitor. Zero custom development.

Transport

TCP/MLLP on hospital medical device VLAN. Same protocol used by every bedside monitor at Northwell.

Routing

Capsule receives → forwards to Health Connect → Health Connect transforms and routes to EHR flowsheet.

Frequency

q15min summaries from continuous stream.
Alert-level changes sent immediately. 24h local buffer on edge box.

Appendix – FHIR R4 Integration (Phase 2–3)

FHIR R4 Observation Resource

Posted directly to Health Connect's FHIR endpoint. Stored in IRIS repository. Queryable by SMART on FHIR widget.

Appendix – Regulatory Decision Table

FDA Regulatory Analysis

QUESTION	ANSWER	SOURCE
Does the AI layer need clearance?	Yes — processes medical images (thermal), outputs clinical score. CDS exemption fails Criterion 1.	FDA CDS Guidance (Jan 2026)
What pathway?	De Novo likely; 510(k) possible using Sepsis ImmunoScore (DEN230036, April 2024) as predicate.	FDA De Novo database
Is Phase 1 software a device?	No — display/store/transfer only. MDDS enforcement discretion applies.	FDA MDDS Guidance
Is Phase 2 trending a device?	Gray area — FDA enforcement discretion covers "historical trending of vital signs." No automated alerts from software.	FDA Enforcement Discretion Examples
Can physicians use camera off-label?	Yes — FDA does not regulate the practice of medicine. 21 CFR 801.4 applies to manufacturers only.	FD&C Act; 21 CFR 801.4
Can hospital configure BPA?	Yes — hospital-configured alert rules are clinical practice, not manufacturer claims. No FDA clearance needed.	CDS Guidance; TTI Warning Letter
Is thermal image data PHI?	Yes — identifiable patient images = PHI. BAA required before any data flows.	HIPAA; HHS guidance
Quality system required?	QMSR (effective Feb 2, 2026). Design controls, CAPA, MDR reporting. PCCP recommended for model updates.	21 CFR 820 (rev. 2026)
Can it be sold as "wellness"?	No — NICU use is inherently clinical. Neonatal monitoring = medical management, not wellness.	FDA General Wellness Guidance (2026)

Appendix – Regulatory Decision Table

FDA Regulatory Analysis

CLAIM	EVIDENCE	SOURCE
CPTD >2°C = sepsis marker	Leante–Castellanos (2012, 2017): Prospective study of 129 VLBW infants. Adjusted OR 23.60, sensitivity 83%, NPV 94%. First clinical sign in 71% of cases.	Pediatric Research
Automated CPTD via camera	Zhang et al. (2025): 40 preterm infants at Nanfang Hospital. MAE 0.297°C for CPTD measurement. Fully automatic, continuous monitoring.	Biomed Optics Express
Polyethylene IR window	92–94% LWIR transmission through LDPE film. Used by Aachen NIRT group and Lorato et al. for multi-camera NICU monitoring.	RWTH Aachen
Zero clinical adoption	"Non-contact technologies have yet to be adopted into clinical practice." 60 studies reviewed, all research/POC stage.	Frontiers Pediatrics (2024)
Camera FDA clearance	Product code LHQ, 21 CFR 884.2980(a). Class I. 14+ cleared devices including FLIR (K033967, 2004).	FDA PCD
AI sepsis predicate exists	Sepsis ImmunoScore (DEN230036, April 2024). First AI sepsis prediction tool authorized via De Novo. Creates predicate pathway.	FDA De Novo
Off-label use legal	FDA does not regulate practice of medicine. 78.9% of NICU devices used off-label (Shah et al.). Standard of care in neonatal medicine.	21 CFR 801.4
TTI cautionary precedent	FDA warning letter (Feb 2019) to Total Thermal Imaging for bundling software with cleared FLIR camera. Camera clearance does not extend to software bundle.	FDA Warning Letter

Appendix – System Architecture

Required

BAA

Thermal images = PHI. 60-day breach reporting.
Up to \$1.9M per violation category.

Required

IRB Protocol

Feinstein Institutes. Dual parental consent. NY
Public Health Law Article 24-A.

Required

IT Security

SOC 2 Type II. HECVAT. Pen test. Possibly
HITRUST CSF for production.

Required

MSA / SaaS

License, SLAs (99.9% uptime), IP ownership,
liability caps, damages waiver.

Required

DUA

If using pilot data for model training. Expert
Determination de-identification (not Safe Harbor).

Required

Clinical Engineering

Biomed validates IEC 60601-1. Camera mount
review. Capsule device type config.

Insurance

E&O + Cyber

\$2–5M technology E&O. \$2–5M cyber liability.
Required by Northwell for any clinical vendor.

Insurance

Product Liability

Separate coverage. "Adjunctive use only" labeling is
the primary liability shield.

Key Framing

Adjunctive Tool

NeoTherm does not replace clinical judgment.
"Adjunctive use" labeling is both FDA-required
and the primary liability shield.

Appendix – Sepsis Alliance

- According a decades long review of cases in the US Sepsis Alliance, backed by the University of Florida, estimates that at least 18 children die each day of sepsis in the United States alone.
- The cost of hospitalizing children for sepsis **increased 25%** between 2005 and 2016, after adjusting for inflation.
- The average length of stay for sepsis patients **is 31.5 days**.

Our Team

Team 218 - Global-Therma:

- Name 1 :
- Name 2 : X X / Role
- Name 3 : X X / Role
- Name 4 : X X / Role
- Name 5 : X X / Role
- Name 6 : X X / Role

Calculating Cost Reduction: Our Methodology



